INCH-POUND

MIL-S-8805/99F(USAF)
7 May 1993
SUPERSEDING
MIL-S-8805/99E(USAF)
19 August 1988

MILITARY SPECIFICATION SHEET

SWITCHES, PUSH, 2 CIRCUIT AND DPDT,
ALTERNATE AND MOMENTARY ACTION, 2 AMPERES, WATERTIGHT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

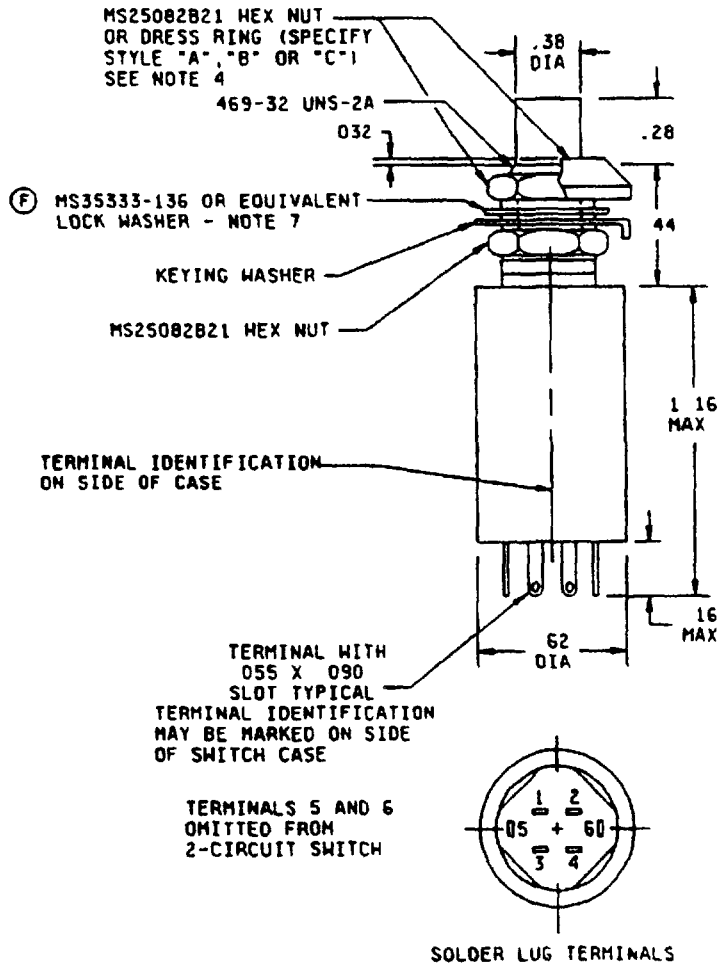
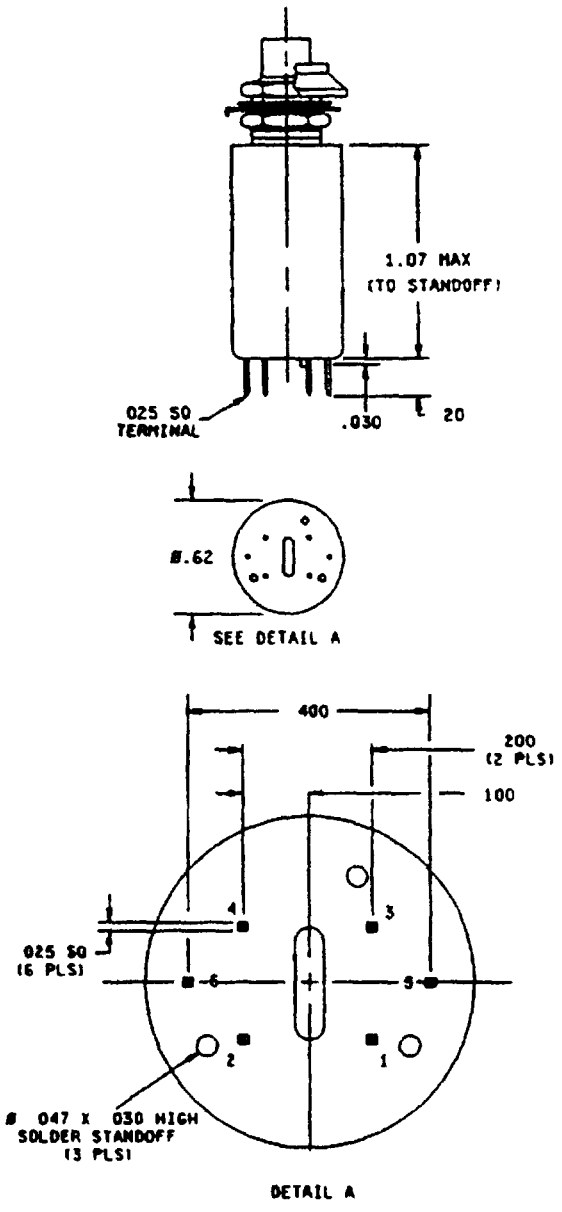


FIGURE 1. Dimensions and configuration.

Ⓕ denotes changes

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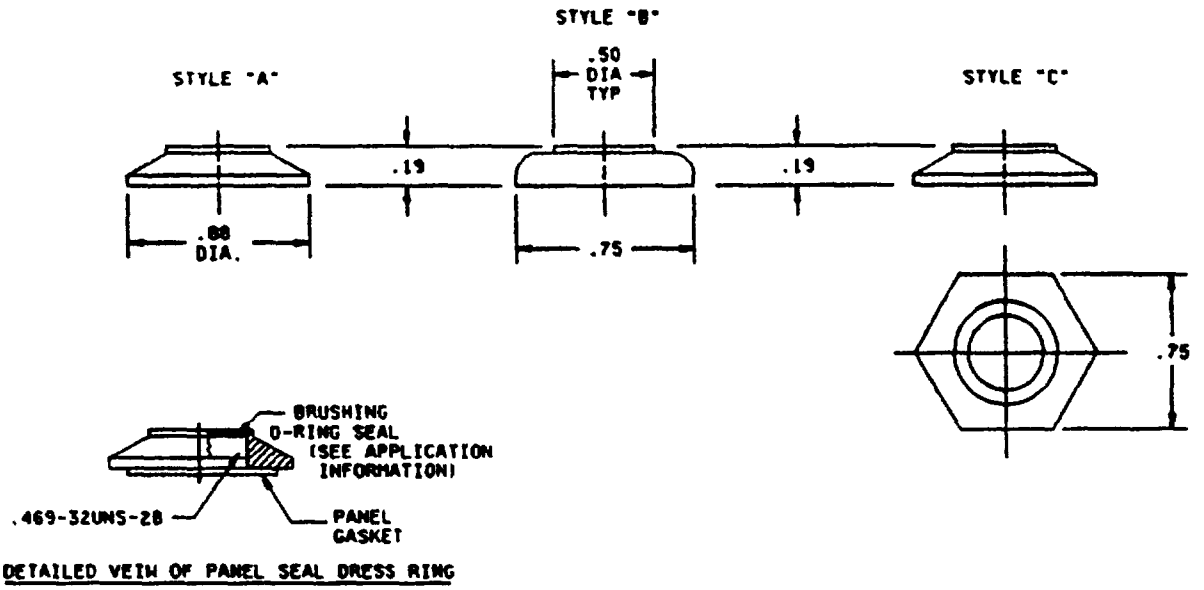


Inches	mm
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.006	0.15
.015	0.38
.016	0.41
.02	0.5
.025	0.63
.030	0.76
.032	0.81
.047	1.19
.048	1.22
.050	1.27
.055	1.40
.090	2.29
.10	2.0
.110	2.79
.136	3.45
.16	4.1
.19	4.8
.20	5.1
.28	7.1
.349	8.86
.375	9.52
.38	9.7
.40	10.0
.44	11.2
.442	11.23
.446	11.33
.468	11.89
.469	11.91
.470	11.94
.50	12.7
.62	15.7
.719	18.26
.75	19.0
.88	22.4
1.07	27.2
1.16	29.5

PRINTED CIRCUIT BOARD TERMINAL

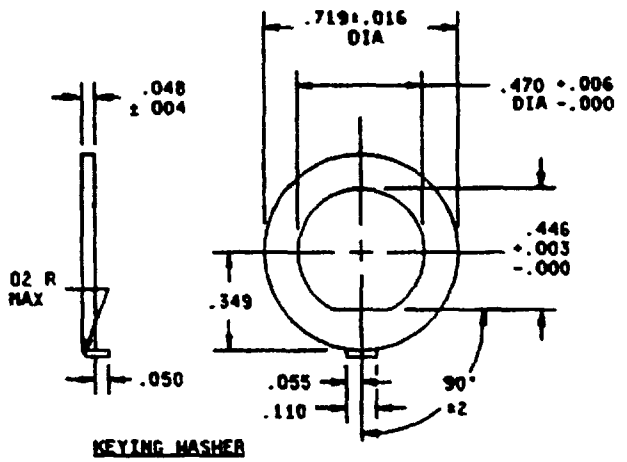
FIGURE 1. Dimensions and configuration - Continued

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DETAILED VEIW OF PANEL SEAL DRESS RING

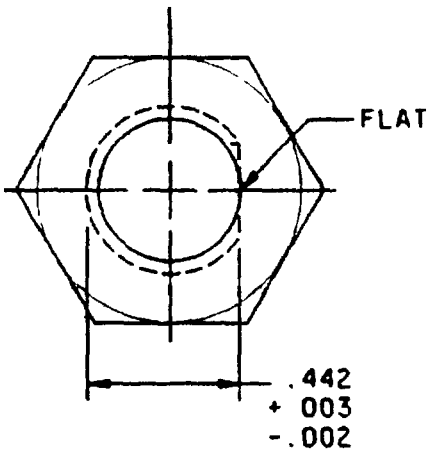
PANEL SEAL DRESS RING



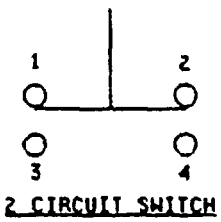
KEYING WASHER

FIGURE 1. Dimensions and configuration - Continued.

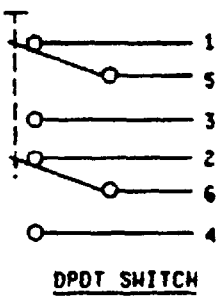
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TOP VIEW

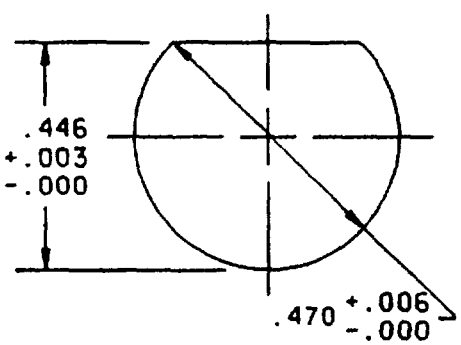


2 CIRCUIT SWITCH

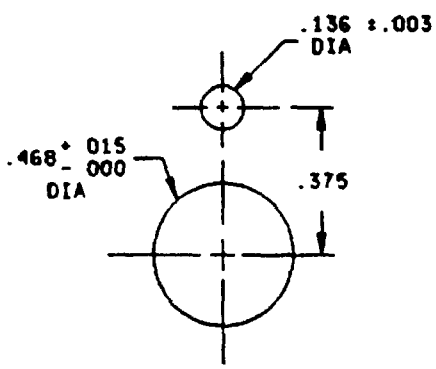


DPDT SWITCH

CIRCUIT SCHEMATIC



PANEL MOUNTING HOLE DIMENSIONS



ALTERNATE MOUNTING HOLE FOR NONTURN WASHER (SEE NOTE 5)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.8 mm) for two place decimals and ±.005 (0.13 mm) for three place decimals.
4. Panel dress ring replaces one hex nut.
5. Alternate mounting hole (2 hole) should not be used for panel seal applications
6. Terminals 5 and 6 omitted from 2-circuit switch.
(F) 7. Functionally equivalent lock washers, as approved by the qualifying activity, may be provided.

FIGURE 1. Dimensions and configuration - Continued

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REQUIREMENTS

Dimensions and configuration: See figure 1.

Enclosure design:

3 (watertight).
6 (splash proof) when panel seal dress ring is specified.

Temperature characteristic: 1 (-55°C to +85°C).

Shock type:

Type M, except 50 g's, test condition A, method 213 of MIL-STD-202: Applicable to all switches.

(F) Type H (high impact), except maximum allowable opening of closed contacts or closing of open contacts is 39 milliseconds: Applicable to momentary action switches only.

Sinusoidal vibration grade: 2 (10 to 2,000 Hz).

Operating characteristics:

(F) Actuating force: 3 to 6 pounds.

Plunger travel: 176 inch maximum.

Materials:

Plunger cap: High temperature polycarbonate.

Contacts: Gold plating, .00003 to .00010 inch thickness.

Terminals: Gold plating, when used, .00005 inch maximum thickness.

Finish:

Housing and dress ring: Aluminum alloy, anodized black in accordance with MIL-A-8625, type II.

Dielectric withstanding voltage:

Sea level: 1,000 V rms.

Altitude: 500 V rms.

Terminal strength

Printed circuit board compatible pin terminals: Applied force shall be 3 pounds minimum.

Mechanical endurance: 50,000 cycles.

(F) Low temperature: Cycling rate of 30 to 65 cycles per minute.

Electrical endurance: 25,000 cycles.

Electrical ratings: See table I.

Qualification inspection (group submission): See table II

Part or Identifying Number (PIN): M8805/99- (dash number from table III).

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Application information: Bushing O-ring seal (applicable to units with panel seal dress ring): The O-ring seal has the potential of stretching and rolling-over onto the flat "D" section of the threaded mounting bushing as the dress bezel ring is installed, causing an improper seating of the o-ring. It is recommended that silicone grease be applied to the o-ring seal in the dress bezel prior to installation to prevent this problem.

TABLE I. Electrical ratings.

Load	Sea level 28 V dc 115 V ac, 400 Hz	70,000 feet 28 V dc
	Amperes	Amperes
Resistive	2.0	2.0
Inductive	1.5	1.5
Lamp	0.5	0.5

TABLE II. Qualification inspection (group submission).

Inspection	Basic switch	Other switch samples	Extent of approval
Qualification inspection table of MIL-S-8805, groups I through VIII inclusive.	M8805/99-029 (All sample units)	Two sample units each of M8805/99-006, -015, -020: Visual and mechanical Vibration Shock Operating characteristics Two sample units of M8805/99-061: Visual and mechanical Solderability Terminal strength Operating characteristics	All

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TABLE III. PIN and characteristics.

PIN M8805/99-	Dress ring	Switch action	Switch circuit	Button color	Terminals
001	A	Alternate action	2 circuit	Black	Solder lug
002	B	Alternate action	2 circuit	Black	Solder lug
003	C	Alternate action	2 circuit	Black	Solder lug
004	None	Alternate action	2 circuit	Black	Solder lug
005	A	Alternate action	2 circuit	Red	Solder lug
006	B	Alternate action	2 circuit	Red	Solder lug
007	C	Alternate action	2 circuit	Red	Solder lug
008	None	Alternate action	2 circuit	Red	Solder lug
033	A	Alternate action	2 circuit	White	Solder lug
034	B	Alternate action	2 circuit	White	Solder lug
035	C	Alternate action	2 circuit	White	Solder lug
036	None	Alternate action	2 circuit	White	Solder lug
009	A	Momentary	2 circuit	Black	Solder lug
010	B	Momentary	2 circuit	Black	Solder lug
011	C	Momentary	2 circuit	Black	Solder lug
012	None	Momentary	2 circuit	Black	Solder lug
013	A	Momentary	2 circuit	Red	Solder lug
014	B	Momentary	2 circuit	Red	Solder lug
015	C	Momentary	2 circuit	Red	Solder lug
016	None	Momentary	2 circuit	Red	Solder lug
037	A	Momentary	2 circuit	White	Solder lug
038	B	Momentary	2 circuit	White	Solder lug
039	C	Momentary	2 circuit	White	Solder lug
040	None	Momentary	2 circuit	White	Solder lug
017	A	Alternate action	DPDT	Black	Solder lug
018	B	Alternate action	DPDT	Black	Solder lug
019	C	Alternate action	DPDT	Black	Solder lug
020	None	Alternate action	DPDT	Black	Solder lug
021	A	Alternate action	DPDT	Red	Solder lug
022	B	Alternate action	DPDT	Red	Solder lug
023	C	Alternate action	DPDT	Red	Solder lug
024	None	Alternate action	DPDT	Red	Solder lug
041	A	Alternate action	DPDT	White	Solder lug
042	B	Alternate action	DPDT	White	Solder lug
043	C	Alternate action	DPDT	White	Solder lug
044	None	Alternate action	DPDT	White	Solder lug
025	A	Momentary	DPDT	Black	Solder lug
026	B	Momentary	DPDT	Black	Solder lug
027	C	Momentary	DPDT	Black	Solder lug
028	None	Momentary	DPDT	Black	Solder lug
029	A	Momentary	DPDT	Red	Solder lug
030	B	Momentary	DPDT	Red	Solder lug
031	C	Momentary	DPDT	Red	Solder lug
032	None	Momentary	DPDT	Red	Solder lug
045	A	Momentary	DPDT	White	Solder lug
046	B	Momentary	DPDT	White	Solder lug
047	C	Momentary	DPDT	White	Solder lug
048	None	Momentary	DPDT	White	Solder lug

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TABLE III PIN and characteristics - Continued

PIN M8805/99-	Dress ring	Switch action	Switch circuit	Button color	Terminals
049	A	Alternate action	2 circuit	Black	Printed circuit
050	B	Alternate action	2 circuit	Black	Printed circuit
051	C	Alternate action	2 circuit	Black	Printed circuit
052	None	Alternate action	2 circuit	Black	Printed circuit
053	A	Alternate action	2 circuit	Red	Printed circuit
054	B	Alternate action	2 circuit	Red	Printed circuit
055	C	Alternate action	2 circuit	Red	Printed circuit
056	None	Alternate action	2 circuit	Red	Printed circuit
057	A	Alternate action	2 circuit	White	Printed circuit
058	B	Alternate action	2 circuit	White	Printed circuit
059	C	Alternate action	2 circuit	White	Printed circuit
060	None	Alternate action	2 circuit	White	Printed circuit
061	A	Momentary	2 circuit	Black	Printed circuit
062	B	Momentary	2 circuit	Black	Printed circuit
063	C	Momentary	2 circuit	Black	Printed circuit
064	None	Momentary	2 circuit	Black	Printed circuit
065	A	Momentary	2 circuit	Red	Printed circuit
066	B	Momentary	2 circuit	Red	Printed circuit
067	C	Momentary	2 circuit	Red	Printed circuit
068	None	Momentary	2 circuit	Red	Printed circuit
069	A	Momentary	2 circuit	White	Printed circuit
070	B	Momentary	2 circuit	White	Printed circuit
071	C	Momentary	2 circuit	White	Printed circuit
072	None	Momentary	2 circuit	White	Printed circuit
073	A	Alternate action	DPDT	Black	Printed circuit
074	B	Alternate action	DPDT	Black	Printed circuit
075	C	Alternate action	DPDT	Black	Printed circuit
076	None	Alternate action	DPDT	Black	Printed circuit
077	A	Alternate action	DPDT	Red	Printed circuit
078	B	Alternate action	DPDT	Red	Printed circuit
079	C	Alternate action	DPDT	Red	Printed circuit
080	None	Alternate action	DPDT	Red	Printed circuit
081	A	Alternate action	DPDT	White	Printed circuit
082	B	Alternate action	DPDT	White	Printed circuit
083	C	Alternate action	DPDT	White	Printed circuit
084	None	Alternate action	DPDT	White	Printed circuit
085	A	Momentary	DPDT	Black	Printed circuit
086	B	Momentary	DPDT	Black	Printed circuit
087	C	Momentary	DPDT	Black	Printed circuit
088	None	Momentary	DPDT	Black	Printed circuit
089	A	Momentary	DPDT	Red	Printed circuit
090	B	Momentary	DPDT	Red	Printed circuit
091	C	Momentary	DPDT	Red	Printed circuit
092	None	Momentary	DPDT	Red	Printed circuit
093	A	Momentary	DPDT	White	Printed circuit
094	B	Momentary	DPDT	White	Printed circuit
095	C	Momentary	DPDT	White	Printed circuit
096	None	Momentary	DPDT	White	Printed circuit

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CONCLUDING MATERIAL

Custodian.
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5930-F731)

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